

Sewon Park

AI-native Platform & Product Engineering Lead

Engineer with roughly eight years of experience connecting physical devices, enterprise software, developer platforms, and production-grade AI verification. I work where device packets, APIs, agent workflows, and release confidence meet.

FOCUS	DIFFERENTIATOR	OPERATING MODEL
AI platform · DevEx · industrial software	Device boundary to developer abstraction	Verification before trust

01

Selected impact

2026 · PRODUCTIZING

Product owner · architecture · full-stack

Developer platform with natural-language verification

Built an integrated Developer platform spanning documentation, an API source of truth, a device emulator, MCP tools, a Linux/AWS sandbox, and evaluation gates. Designed natural-language recall and replay for complex user, door, authentication, and event scenario history. Built independently with an AI-native workflow and now being hardened for product release.

- One operation model shared by six developer-facing surfaces
- Same gateway concept from device emulator toward physical-device workflows
- Recognition through a company-wide engineering excellence program

2026 · IN PROGRESS

Legacy modernization with meaning-preservation gates

Leading an AI-native frontend rewrite from a mixed Angular/React architecture. Backend payloads and responses, API contracts, React behavior, useEffect, DOM, UI, CSS assertions, and E2E flows are verified as separate release gates.

2025 · DEPLOYED

Global Technical Support chatbot

Developed the frontend and backend for an internal Technical Support chatbot used by overseas offices and employees, and deployed the service on AWS EC2.

02

Experience

2021.02—
PRESENT

Platform & Product Engineering · AI-native Mission Lead

Suprema · Seongnam, Korea

- Lead cross-functional AI-native missions covering developer tooling, product modernization, and version-aware E2E verification.
- Developed vertically across device packets and SDKs, user and identity domains, distributed events, backend services, web products, and cloud integrations.
- Designed reusable AI verification and automated testing workflows, and shared the operating method through mentoring and internal technical sessions.

2020.04—2021.01

Research Engineer · Sensor Software

HBrain · Korea

- Developed Windows software for radar, camera, and FPGA-based pedestrian-detection systems using C/C++, MFC, OpenGL, and device SDKs.
- Contributed to public R&D projects and wrote requirements, protocol, and operational-scenario documents.

2018.10—2020.02

Software Developer · Network Management

AtData · Korea

- Developed and maintained web applications for SK Telecom's network-management program, including IP configuration, transmission views, and raw-data processing.
- Implemented Jenkins-based deployment automation and supported new server environments.

Research & education

RESEARCH IN PROGRESS

Explainable retrieval & hybrid model evaluation

- Expert-knowledge and VLM-based explainable traditional-rag retrieval and recommendation
- Hybrid quantum-classical framework comparing classical ML and QML candidates under shared data splits, budgets, and evaluation gates

2026—PRESENT

Ph.D. coursework, AI Convergence

Sungkyunkwan University

2019—2021

M.S., Applied Data Science

Sungkyunkwan University

Thesis: response-target prediction for public petitions under class imbalance

2010—2017

B.S., Maritime Police Systems

Gyeongsang National University

Technical foundation

Platform & product

Java · Kotlin · Python · JavaScript/TypeScript · Spring · React · REST · gRPC

Device & domain

C/C++ · device SDKs · packet debugging · biometric authentication · user/door/auth/event flows

AI & evaluation

LLM · MCP · RAG · agent workflows · PyTorch · contract/unit/DOM/UI/CSS/E2E testing

Data & infrastructure

MariaDB · Microsoft SQL Server · AWS Lambda/S3/EC2 · Docker/Linux · Jenkins · GitLab CI/CD

Engineering enablement

Led intern projects for LLM-assisted test generation, mentored AI learners, and shared verification-first practices through internal presentations and examples.

Selected presentation

Development of AI traffic-safety element technology and an integrated system — Korean Society of Intelligent Transport Systems, oral presentation.